

MTH209: Worksheet 3

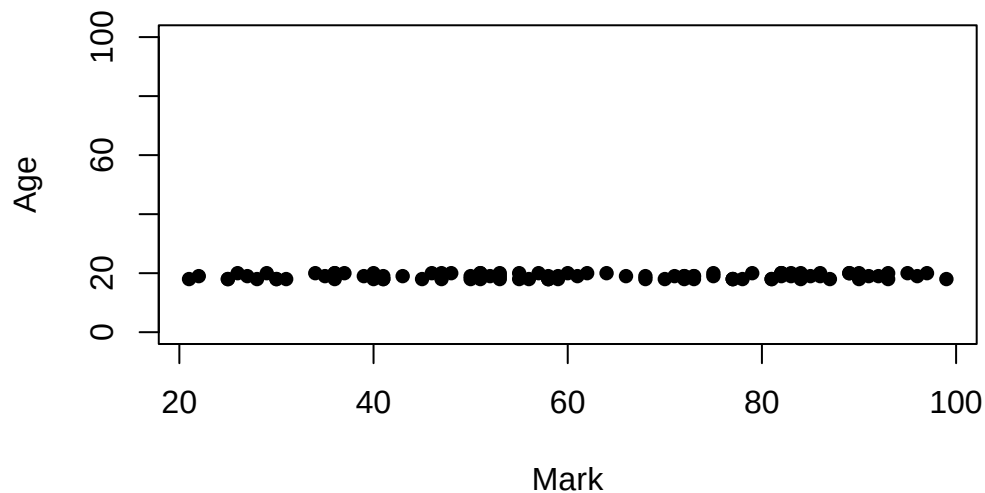
Dimension Reduction

Simple Problem

Consider a dataset where we have collected the following information:

- the marks of BS students in MTH209 Midsem
- the age of the student

The dataset might look something like this:



A few things are immediately noticeable.

1. There is significant difference in marks of the students in the exam.
2. There is very little difference in the ages of the students taking the exam.

Suppose you are in a situation now where the space on your machine is running low and you cannot store both these variables. You must choose only one variable to keep, and discard the other. The act of discarding one component is the most simple form of *dimension reduction*.

Question: Which of the two variables would you discard?

Hopefully it is clear that we would like retain marks as it carries more information and discard age, because it carries less information. This brings us to another question:

Question: How, in this case, do we quantify information?

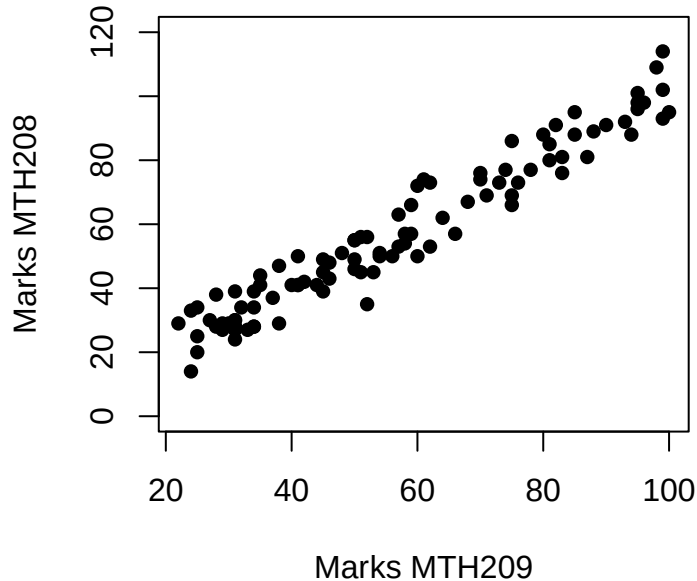
A possible answer for this question is the variance in the marks variable is much larger than the variance in the age variable. A larger variance allows us to separate and identify individual data points.

Less Simpler Problem

Now consider a dataset where we have collected the following information:

- the marks of BS students in MTH209 Midsem
- the marks of BS students in MTH208 Endsem

The dataset might look something like this:

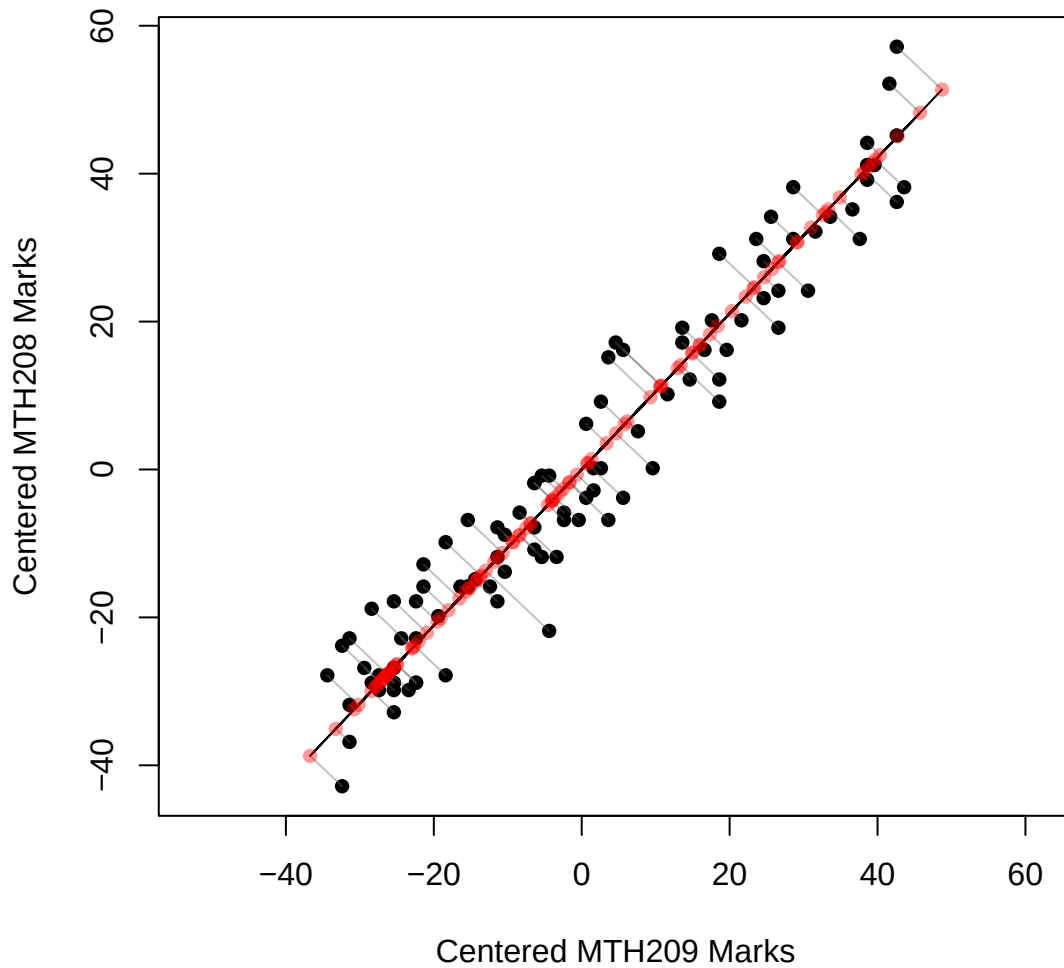


Here again, suppose our machine runs out of memory and we need to do dimension reduction. Which of the two should I remove? Is there a clear answer?

Both the variables display similar variability so it is not clear which variable is useless. However, given that one's MTH209 marks gives us some information about MTH208 marks (there is some correlation), one wonders if we can use this somehow?

Projecting on to a line

Consider the following line in the plot (where we have centered the observations around 0 on both axis).



What if I projected every point on to this line (the light red points), then all points line on a 1-dimensional subspace of the full space. That means, dimension reduction can happen!

Mathematically, let each observation be a d -dimensional vector $x_i = (x_{i1}, x_{i2}, \dots, x_{id})$. We have a vector v such that each data point x_i when projected on to that vector is:

$$(v^\top x_i)v.$$

and the length of each projection. is $v^\top x_i$. Now we can extend the understanding of variability to this. Can we find the vector v such that

$$\text{Var}(v^\top x_i)$$

is maximized. Now $x_i \in \mathbb{R}^d$ is our random quantity from the data. Suppose it has a covariance matrix Σ , then

$$\text{Var}(v^\top x_i) = v^\top \Sigma v$$

So we have to find a vector such that

$$\arg \max_{v: \|v\|=1} v^\top \Sigma v.$$

As it turns out, this is the largest eigenvector of Σ . That means, instead of saving each x_i , I can save $v^\top x_i$, reducing the number of dimensions by 1!

Principal Component Analysis

Understanding this motivation, we can set up a general idea. For a given dataset $x_1, x_2, \dots, x_n$ with each $x_i \in \mathbb{R}^d$, and such that $\text{Var}(x_i) = \Sigma$, principal components analysis does the following:

1. Estimate Σ from the data:

$$\widehat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)(x_i - \bar{x}_n)^\top.$$

2. Find the d eigenvectors of $\widehat{\Sigma}$: $v_1, v_2, \dots, v_d$. These are called the principal components and are essentially linear combinations of the d dimensions.
3. Choose a $k < d$ and obtain for each i :

$$v_1^\top x_i, v_2^\top x_i, \dots, v_k^\top x_i.$$

This completes dimension reduction via Principal Components Analysis. The vectors v_i are the principal components and $v_1^\top x_i, v_2^\top x_i, \dots, v_k^\top x_i$ are the k -dimensional scores.

Note that if we make a matrix $V_k = [v_1 \ v_2 \ \dots \ v_k] \in \mathbb{R}^{n \times k}$, and the original data is designed in $X \in \mathbb{R}^{n \times d}$ such that each x_i is a row, then: XV_k are the scores and $XV_k V_k^\top$ is the reconstructed data lying in a lower dimensional space.

Questions

1. (Homework) Try to recreate the plots I made above by yourself.
2. Consider the popular `iris` dataset in R.

```
data(iris)
plot(iris)
plot(iris, col = iris$Species)
```

We will manually do PCA for this dataset (for the first four columns). Let's start:

- a. Calculate the sample covariance matrix $\widehat{\Sigma}$ using `cov` function.
 - b. Using `eigen` function, obtain the 4 eigenvectors of $\widehat{\Sigma}$.
 - c. Choose $k = 2$ eigenvectors and obtain the principal components and the scores.
 - d. Verify your answer with the `prcomp` function in R.
 - e. Obtain the restructured data in the 4-dimensional space and calculate the eigenvalues of the sample covariance matrix. What do you observe and why?
3. In the file `pca_plots.R` is the function `plot_loadings()` that make a nice plot of the principal component weights. Source this code by using `source(pca_plots.R)`.

4. Using this plot, can you understand what are the features of the first two principal components of the previous example?
5. A key question is choosing the number of principal components. This is done often by the screeplot. Run `screeplot()` on the output of `prcomp`. This plots the eigenvalues of the $\widehat{\Sigma}$. How would you use this plot to choose k ?
6. A dataset wine is provided to you in a txt file. More information is available here: <https://archive.ics.uci.edu/dataset/109/wine>
 - a. Load this data and implement PCA, removing the first column of the kind of wine. What do you observe? Can we fix this?
 - b. Scale the data using function `scale`.
 - c. Apply PCA again. Now what do you observe?
 - d. Plot the first two PC score components, and color the points based on the Class of wine. What do you observe?
 - e. What would be the choice of k ?
7. Load the grocery score data in `my_basket.csv`. This is the data from: <https://bradleyboehmke.github.io/HOML/pca.html>

“This data set identifies items and quantities purchased for 2,000 transactions from a grocery store. The objective is to identify common groupings of items purchased together.”

 - a. Plot the PCA loadings. Do you see any theme for the first principal component?
 - b. Make a two variable PC plot using `plot_twoPC` function available in `pca_plots.R`
 - c. Choose an appropriate number of principal components.